

AUTOMATIC INDICATOR SYSTEM USING ESP32 AND GPS

A MINI PROJECT REPORT

Submitted by

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under the supervision of

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DECLARATION

We hereby declare that this submission is our own original work. To the best of our knowledge and belief, it contains no material previously published or written by another individual, nor any material that has been submitted for the award of any other degree or diploma at any university or institution of higher learning, except where proper acknowledgment is provided in the text.

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CERTIFICATE

This is to certify that the report entitled "**AUTOMATIC INDICATOR SYSTEM USING ESP32 AND GPS**", submitted by **MOHAMMED IRFAN B (ATP23EC029)**, **PRASIN KRISHNA E (ATP23EC038)**, **SAINUL ABIDH PS (ATP23EC044)** and **SREE VISHNU S (ATP23EC049)** to APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Bachelor of Technology degree in Electronics and Communication Engineering, is a bona fide record of the project work related to the course (ECD 334) **MINI PROJECT**, carried out under our guidance and supervision. This report has not been submitted in any form to any other university or institution for any purpose.

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ABSTRACT

The GPS Based Automatic Turn Indicator System is designed to automatically detect vehicle turning direction and activate the corresponding turn indicator without manual intervention. The system uses an ESP32 microcontroller along with a GPS module to continuously monitor the vehicle's geographical position and heading direction. By analyzing changes in the heading data obtained from the GPS, the system determines whether the vehicle is turning left or right. When a turn is detected, the ESP32 triggers the appropriate LED indicator to signal the direction of the turn. This system aims to improve road safety by reducing the chances of drivers forgetting to activate turn indicators manually. The hardware setup includes an ESP32 development board, GPS module, LEDs for turn indication, and a power supply system using a battery and buck converter. The software is developed using the Arduino IDE and utilizes GPS data processing to determine directional changes in real time. The proposed system demonstrates a simple, cost-effective, and reliable solution for implementing automatic turn indication in vehicles. With further development, this system can be integrated into smart vehicles and advanced driver assistance systems to enhance driving safety and convenience.

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ABBREVIATIONS

Abbreviation	Expansion
GPS	Global Positioning System
ESP32	Espressif 32-bit Microcontroller
LED	Light Emitting Diode
DC	Direct Current
AC	Alternating Current
WiFi	Wireless Fidelity
PCB	Printed Circuit Board
IC	Integrated Circuit
ADAS	Advanced Driver Assistance System

Chapter 1

INTRODUCTION

1.1 Socio-Economic Relevance Of The Project

Road safety has become a major concern worldwide due to the increasing number of vehicles on the road. A significant percentage of road accidents occur due to human errors such as failure to use turn indicators while changing direction. When drivers forget or neglect to activate turn signals, it can lead to confusion among other road users and may result in collisions or unsafe driving situations. Such accidents not only cause injuries and loss of life but also lead to economic losses due to vehicle damage, medical expenses, and traffic disruptions.

The GPS Based Automatic Turn Indicator System addresses this issue by automating the process of turn indication using modern embedded technologies. The system utilizes a GPS module and an ESP32 microcontroller to continuously monitor the vehicle's direction and detect turning movements. When a change in direction is detected, the system automatically activates the corresponding turn indicator, thereby reducing the dependence on manual driver actions.

From a socio-economic perspective, the implementation of such an automated system can contribute to improved road safety and reduced accident rates. It also helps in minimizing human errors and enhances the overall driving experience. Furthermore, the development of low-cost intelligent systems like this project promotes the adoption of smart transportation technologies, which can play an important role in building safer and more efficient transportation infrastructure in the future.

1.2 Existing System

In conventional vehicles, turn indicators are manually operated by the driver using a mechanical switch located near the steering wheel. The driver is responsible for activating the indicator before making a turn and turning it off after completing the maneuver. While this system is simple and widely used, it relies entirely on the driver's attention and timely action.

In many situations, drivers forget to activate the turn indicator or fail to turn it off after completing the turn. This can lead to confusion for other drivers and may increase the risk of road accidents. In busy traffic conditions or during long drives, such human errors are common and can compromise road safety.

Some modern vehicles include advanced driver assistance systems that provide automated signaling or navigation-based turn guidance. However, these systems are often expensive and integrated only into high-end vehicles. As a result, they are not accessible for most users, especially in low-cost or older vehicles.

Therefore, there is a need for a simple, cost-effective, and automated solution that can detect vehicle turning direction and activate the corresponding indicator without requiring manual input from the driver. The proposed system aims to address this need by using GPS-based direction detection and a microcontroller-based control system.

1.3 Motivation

With the rapid growth in the number of vehicles on the road, ensuring road safety has become increasingly important. One of the common causes of road accidents is the failure of drivers to properly use turn indicators while changing direction. Drivers may forget to activate the indicator, activate it too late, or forget to turn it off after completing the turn. Such situations can create confusion among other road users and increase the chances of accidents.

The motivation behind this project is to develop a simple and effective system that can automatically detect the turning direction of a vehicle and activate the corresponding turn indicator. By utilizing GPS technology and a microcontroller-based

system, it is possible to monitor the vehicle's direction and identify turning movements without requiring manual intervention from the driver.

This project is also motivated by the growing importance of embedded systems and intelligent transportation technologies in modern vehicles. Developing a low-cost automatic indicator system demonstrates how simple electronic components and sensors can be used to improve driving safety and convenience. The proposed system can serve as a foundation for further advancements in smart vehicle technologies and driver assistance systems.

1.4 Problem Definition

In conventional vehicles, turn indicators are manually operated by the driver. This manual system depends entirely on the driver's awareness and timely action. In many situations, drivers forget to activate the turn indicator before making a turn or fail to turn it off after completing the maneuver. Such mistakes can lead to confusion among other road users and increase the possibility of road accidents.

In heavy traffic conditions or during long-distance driving, these human errors become more frequent. Existing automated solutions are mostly available only in high-end vehicles and are expensive, making them inaccessible for many users. Therefore, there is a need for a simple and cost-effective system that can automatically detect the turning direction of a vehicle and activate the corresponding turn indicator without relying completely on the driver.

The problem addressed in this project is the development of a reliable system that can identify vehicle turning movements using GPS data and automatically trigger the appropriate turn indicator to improve road safety.

1.5 Objectives

The main objective of this project is to design and develop a GPS based automatic turn indicator system that can detect vehicle turning direction and activate the corresponding indicator automatically. The specific objectives of the project include:

- To design a system that can monitor the vehicle's position and direction using a GPS module.
- To process GPS data using an ESP32 microcontroller to detect changes in direction.
- To automatically trigger left or right turn indicators based on the detected turning direction.
- To develop a low-cost and efficient system that can be easily implemented in vehicles.
- To improve road safety by reducing human errors related to manual indicator usage.

1.6 Scope of the Project

The scope of this project focuses on the development of a prototype system that demonstrates automatic turn indication using GPS data and a microcontroller-based control system. The system detects directional changes using GPS heading information and activates the corresponding LED indicators to represent left or right turns. The project mainly emphasizes the design and implementation of the hardware interface and software logic required for automatic turn detection. The developed prototype can be used as a basic model for integrating intelligent signaling systems in vehicles. In the future, the system can be further enhanced by integrating additional sensors, real-time navigation data, and vehicle communication systems to provide more accurate and reliable turn detection. Such improvements can contribute to the development of advanced driver assistance systems and smart transportation technologies.

Chapter 2

LITERATURE REVIEW

In recent years, several studies have explored the use of Global Positioning System (GPS) technology in intelligent transportation systems. According to Sharma et al. (2023), GPS-based vehicle monitoring systems provide accurate location and navigation information that can be used to analyze vehicle movement and direction. The study demonstrated that GPS data such as latitude, longitude, and heading information can be effectively utilized to detect changes in vehicle direction and improve navigation assistance systems[1]

Another study by Kumar and Patel (2024) proposed a microcontroller-based automatic vehicle indicator system designed to reduce the dependency on manual indicator operation. The system uses sensors and embedded controllers to detect vehicle movement and automatically activate the corresponding turn indicators. The research showed that automation of vehicle indicators can significantly reduce driver errors and improve road safety [2]

Research conducted by Zhang et al. (2023) focused on integrating Internet of Things (IoT) technology with vehicle safety systems. In this work, different sensors and communication modules were used to monitor vehicle movement and transmit real-time data for analysis. The study highlighted the importance of embedded systems and IoT in developing intelligent vehicle safety solutions and improving traffic management [3]

Singh et al. (2024) proposed an embedded system-based driver assistance module that integrates GPS data with microcontroller processing to detect vehicle

movement patterns. The system processes heading and directional changes to determine vehicle turning behavior. The results showed that such systems can effectively support automated signaling mechanisms in modern vehicles [4]

Similarly, Ahmed and Hassan (2023) developed a smart vehicle signaling system using GPS and microcontroller technology to automatically activate turn indicators. Their research demonstrated that integrating positioning data with embedded control systems can significantly improve vehicle communication with surrounding traffic and contribute to safer driving environments[5]

Chapter 3

PROPOSED SYSTEM

3.1 Proposed System

The proposed system is a GPS-based automatic turn indicator system designed to detect the turning direction of a vehicle and automatically activate the corresponding turn indicator. The system utilizes a GPS module to continuously monitor the geographical position of the vehicle and determine its heading direction. The collected GPS data is processed by an ESP32 microcontroller to identify changes in direction during vehicle movement.

When the system detects a significant change in the vehicle's heading angle, it determines whether the vehicle is turning left or right. Based on this analysis, the microcontroller activates the appropriate LED indicator to represent the turning direction. This automatic indication helps reduce dependency on manual indicator operation and improves driving safety.

The proposed system is designed using low-cost hardware components including an ESP32 microcontroller, a GPS module, LED indicators, and a power supply system using a buck converter. The entire system is compact, energy efficient, and suitable for implementation in vehicles as a driver assistance feature.

3.1.1 Block Diagrams

The block diagram of the proposed system consists of a power supply unit, ESP32 microcontroller, GPS module, and indicator LEDs. The vehicle battery provides a 12V power supply which is converted to a lower voltage using a buck converter suitable for the ESP32 and other components. The ESP32 microcontroller acts as the central processing unit of the system. It receives location and heading information from

the GPS module through serial communication. Based on the received data, the microcontroller analyzes the vehicle's direction of movement. When a turning movement is detected, the ESP32 activates the corresponding LED indicator. The left LED represents a left turn while the right LED represents a right turn. This automatic activation provides a visual indication of the vehicle's turning direction.

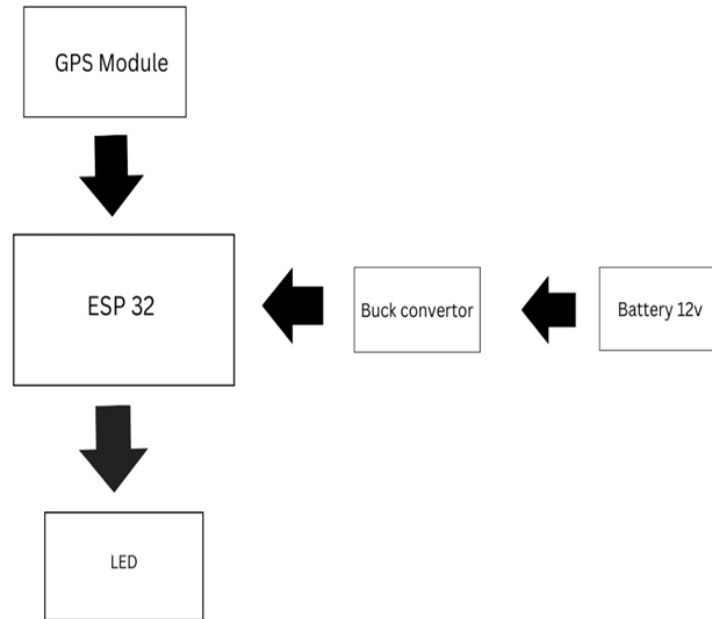


Figure 3.1: Block Diagram

3.1.2 Hardware Components

3.1.2.1 ESP32

The ESP32 is a low-power microcontroller widely used in embedded systems and IoT applications. It features a dual-core processor, built-in Wi-Fi, and Bluetooth connectivity. In the proposed system, the ESP32 acts as the main controller that processes GPS data and determines the turning direction of the vehicle. Based on the calculated heading change, it controls the LED indicators for left and right turns.



Figure 3.2: ESP32 Microcontroller

3.1.2.2 GPS Module

The GPS module is used to obtain real-time location data from satellites. It provides parameters such as latitude, longitude, and heading information. In this project, the GPS module communicates with the ESP32 using serial communication and continuously transmits location data which is used to detect vehicle movement and turning direction.



Figure 3.3: GPS Module

3.1.2.3 Buck Converter

A buck converter is a DC-DC step-down voltage regulator used to reduce the input voltage to a lower stable output voltage. In the proposed system, the buck converter converts the 12V supply from the vehicle battery to a suitable voltage level required by the ESP32 and other electronic components.



Figure 3.4: Buck Converter

3.1.2.4 LED Indicators

LED indicators are used to represent the turning direction of the vehicle. Two LEDs are used to indicate left and right turns respectively. These LEDs are controlled by the ESP32 microcontroller and blink when a turning movement is detected.



Figure 3.5: LED

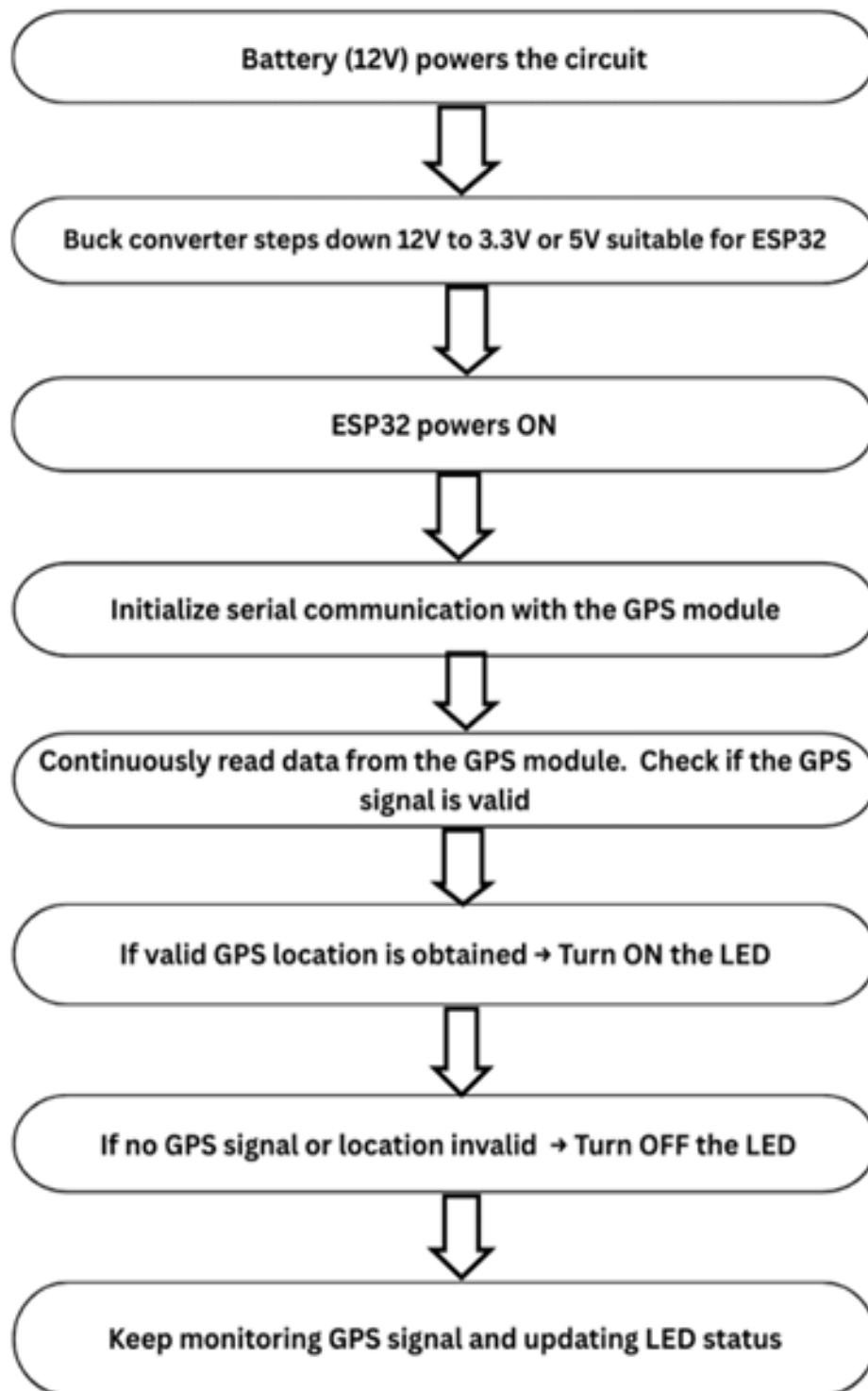
3.1.3 Software Design

The software for the proposed system is developed using the Arduino Integrated Development Environment (Arduino IDE). The ESP32 microcontroller is programmed using embedded C/C++ to read data from the GPS module and control the LED

indicators based on the detected turning direction. The GPS module continuously transmits NMEA data containing information such as latitude, longitude, and course (heading). The ESP32 reads this data through serial communication and extracts the heading information using a GPS parsing library. The program compares the current heading value with the previous heading value to determine whether the vehicle is turning. If the change in heading exceeds a predefined threshold, the system identifies it as a turning movement. Based on the direction of change, the corresponding LED indicator is activated to represent a left or right turn signal.

3.1.4 System Algorithm

The operation of the proposed system follows a simple algorithm for detecting vehicle turning directions using GPS data. The hardware implementation of the system centers on a unified circuit architecture where the ESP32 acts as the primary controller, managing both data input and signaling output. Power is distributed through a 5V rail, which provides stable energy to both the ESP32 and the GPS module, ensuring reliable serial communication. The GPS module is interfaced with the microcontroller via its transmit (TX) and receive (RX) pins, enabling the continuous flow of NMEA data for directional processing. For the output interface, the circuit employs two separate LED indicators representing the left and right signals. Each LED is wired in series with a 220-ohm current-limiting resistor and connected to specific General Purpose Input/Output (GPIO) pins on the ESP32. This configuration prevents overcurrent damage to the LEDs while allowing the microcontroller to execute the pulse-width modulation or digital switching required to simulate a standard vehicle turn signal. This compact and modular wiring scheme allows the entire unit to be easily integrated into a vehicle's existing electrical framework.

**Figure 3.6:** Flow Chart

3.1.5 Working Principle

The working principle of the proposed GPS-based automatic turn indicator system is based on analyzing the directional movement of a vehicle using GPS heading information. The system mainly consists of an ESP32 microcontroller, a GPS module, LED indicators, and a power supply unit. Initially, the system is powered using a 12V power source which is regulated using a buck converter to provide a stable voltage suitable for the ESP32 and GPS module. Once powered, the GPS module begins receiving signals from multiple satellites and determines the geographical position of the device. The GPS module continuously sends location data such as latitude, longitude, and heading direction to the ESP32 microcontroller through serial communication. The ESP32 processes the received GPS data and extracts the heading information, which indicates the direction of movement of the vehicle. By comparing the current heading value with the previously recorded heading value, the system determines whether a significant directional change has occurred. If the heading change corresponds to a left turn, the ESP32 activates the left LED indicator connected to the corresponding GPIO pin. Similarly, if the heading change corresponds to a right turn, the right LED indicator is activated. The LED blinks to represent the turn signal, indicating the turning direction of the vehicle. This automated mechanism allows the system to detect vehicle turning movements without manual intervention and provides appropriate visual indication using LED signals. The continuous monitoring of GPS data enables the system to respond dynamically to changes in vehicle direction, thereby enhancing driving safety and reducing human errors in using manual turn indicators.

Chapter 4

RESULTS AND DISCUSSIONS

- 1. Implementation:** The implementation of the proposed GPS-based automatic turn indicator system was carried out using an ESP32 microcontroller, GPS module, LED indicators, and a buck converter for power regulation. The system was developed by integrating both hardware and software components to achieve automatic turn detection based on GPS data. The GPS module was interfaced with the ESP32 using UART serial communication. The module continuously receives satellite signals and provides real-time data such as latitude, longitude, and heading direction. This data is transmitted to the ESP32 microcontroller for processing. The ESP32 was programmed using the Arduino IDE. The program reads the GPS data and extracts the heading information using appropriate libraries. By comparing the current heading value with the previous heading value, the system determines whether a directional change has occurred. LED indicators were connected to the GPIO pins of the ESP32 to represent left and right turn signals. When a significant change in heading is detected, the ESP32 activates the corresponding LED to indicate the turning direction. The entire system was powered using a 12V source, which was regulated using a buck converter to provide a stable voltage for the components.

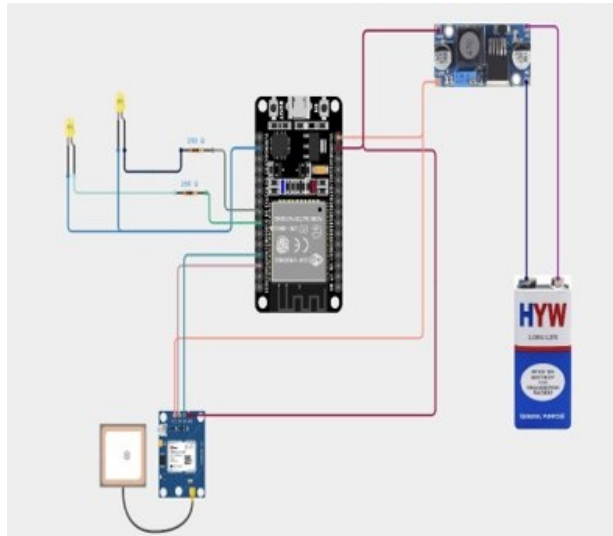


Figure 4.1: Circuit Setup

2. Results and Analysis: The developed system was tested under outdoor conditions to ensure proper GPS signal reception. After powering the system, the GPS module successfully acquired satellite signals and started transmitting location data including latitude, longitude, and heading information. The ESP32 microcontroller continuously processed the heading data and monitored changes in direction. When the system detected a change in heading corresponding to a left turn, the left LED indicator was activated. Similarly, when a right turn was detected, the right LED indicator was activated. The system demonstrated the ability to automatically detect turning movements based on GPS data and provide appropriate visual indication using LEDs. The results confirm that the proposed system can effectively reduce the dependency on manual indicator operation.



Figure 4.2: Left LED Indication

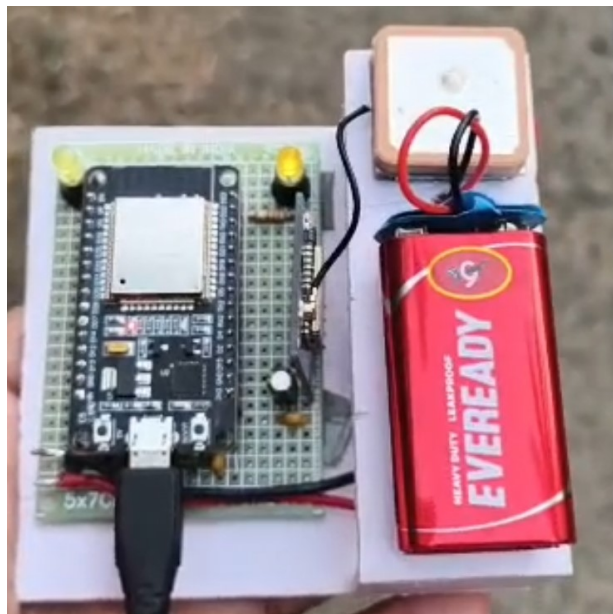


Figure 4.3: Right LED Indication

The main objective of this project is to design and develop a GPS based automatic turn indicator system that can detect vehicle turning direction and activate the corresponding indicator automatically. The specific objectives of the project include:

- To design a system that can monitor the vehicle's position and direction using a GPS module.

- To process GPS data using an ESP32 microcontroller to detect changes in direction.
- To automatically trigger left or right turn indicators based on the detected turning direction.
- To develop a low-cost and efficient system that can be easily implemented in vehicles.
- To improve road safety by reducing human errors related to manual indicator usage.

Discussion:The experimental results show that the proposed system is capable of detecting vehicle turning directions using GPS heading information and automatically activating the corresponding indicators. The system performs well in outdoor environments where GPS signals are strong and stable. However, the system performance is influenced by the accuracy of GPS data. Small or gradual turns may not always be detected accurately due to limitations in heading sensitivity. Additionally, there may be slight delays in response due to the time required for GPS signal processing. Despite these limitations, the system provides a simple and effective solution for automatic turn indication. The use of low-cost components and simple design makes the system suitable for practical implementation and further development.

Code:`include <TinyGPS++.h>
include <HardwareSerial.h> include <WiFi.h>
TinyGPSPlus gps;
HardwareSerial gpsSerial(2);
define LEFT_LED22
define RIGHT_LED23
float prevCourse = 0;
bool firstFix = true;
int turnDirection = 0;
bool blinking = false;`

```
int blinkCount = 0;
unsigned long lastBlinkTime = 0;
bool ledState = false;
void setup()
Serial.begin(115200);
WiFi.mode(WIFI_OFF);
gpsSerial.begin(9600, SERIAL_8N1, 16, 17);
const byte rate2Hz[] = {0xB5, 0x62, 0x06, 0x08, 0x06, 0x00, 0xF4, 0x01, 0x01,
0x00, 0x01, 0x00, 0x0B, 0x77};
for(int i = 0; i < sizeof(rate2Hz); i++) gpsSerial.write(rate2Hz[i]);
pinMode(LEFT_LED, OUTPUT);
pinMode(RIGHT_LED, OUTPUT);
digitalWrite(LEFT_LED, LOW);
digitalWrite(RIGHT_LED, LOW);
Serial.println("SYSTEM READY - OFFLINE LOW POWER MODE");
void startBlinking(int dir)
blinking = true;
blinkCount = 0;
ledState = false;
turnDirection = dir;
lastBlinkTime = millis();
void stopBlinking()
blinking = false;
turnDirection = 0;
digitalWrite(LEFT_LED, LOW);
digitalWrite(RIGHT_LED, LOW);
void loop()
while (gpsSerial.available() > 0)
gps.encode(gpsSerial.read());
if (gps.location.isValid() gps.course.isUpdated())
```

```
float currentSpeed = gps.speed.kmph();
float currentCourse = gps.course.deg();
if (firstFix)
  prevCourse = currentCourse;
  firstFix = false;
else
  float diff = currentCourse - prevCourse;
  if (diff > 180.0) diff -= 360.0;
  if (diff < -180.0) diff += 360.0;
  if (!blinking currentSpeed > 1.0)
    if (diff > 8.0)
      Serial.println("»» RIGHT TURN");
      startBlinking(1);
      prevCourse = currentCourse;
    else if (diff < -8.0)
      Serial.println("«« LEFT TURN");
      startBlinking(-1);
      prevCourse = currentCourse;
    if (!blinking)
      prevCourse = currentCourse;
    if (blinking)
      if (millis() - lastBlinkTime > 300)
        lastBlinkTime = millis();
        ledState = !ledState;
      if (turnDirection == 1)
        digitalWrite(RIGHT_LED, ledState);
        digitalWrite(LEFT_LED, LOW);
      else if (turnDirection == -1)
        digitalWrite(LEFT_LED, ledState);
        digitalWrite(RIGHT_LED, LOW);
```

```
if (ledState == HIGH)
  blinkCount++;
if (blinkCount >= 10)
  stopBlinking();
```

Chapter 5

FUTURE SCOPE

The proposed GPS-based automatic turn indicator system demonstrates a simple and effective approach for automating vehicle signaling. However, there is significant scope for further improvement and enhancement to make the system more accurate, reliable, and suitable for real-world applications. One possible improvement is the integration of additional sensors such as gyroscopes and accelerometers along with the GPS module. These sensors can provide more precise information about vehicle motion and orientation, thereby improving the accuracy of turn detection, especially in situations where GPS signals are weak or unavailable. Another important enhancement is the integration of real-time navigation systems. By using map-based navigation data, the system can predict upcoming turns in advance and activate indicators before the vehicle actually starts turning. This would make the system more intelligent and closer to modern driver assistance systems. The system can also be extended by incorporating Internet of Things (IoT) technology. With IoT connectivity, the vehicle's movement and signaling status can be monitored remotely using mobile applications. This can be useful for vehicle tracking, fleet management, and safety monitoring. Further improvements can include integrating the system directly with the vehicle's existing electrical system instead of using external LEDs. This would allow the automatic control of actual vehicle indicators, making the system more practical for real-world implementation. In addition, the use of artificial intelligence and machine learning algorithms can enhance the system by predicting driver behavior and improving decision-making accuracy. This can lead to the development of more advanced driver assistance systems.

Chapter 6

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